MAIS PAC

IPhO 04.02: Vectors and Complex Numbers · Worked Solutions

Vectors and Complex Numbers: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Components and a resultant. Return →

(a) 50 {N}: (50cos 30°, 50sin 30°) = (43.3, 25.0). 40 {N}: (40cos 120°, 40sin 120°) = (-20.0, 34.6).

(b) Sum: (23.3, 59.6), magnitude √(543 + 3552) = 64.0 {N} at tan ^{-1}(59.6/23.3) = 68.6° above the horizontal.

(c) Because the two forces are 90° apart, the resultant is √(50²+40²) = 64.0 {N} — the diagonal of the rectangle, always shorter than the sum of the sides. **Check:** the angle between the forces is 120°-30° = 90°, which is exactly the case where Pythagoras applies.

Solution 2 — Straight-line motion in two dimensions. Return →

(a) The velocity components are 4.0 across and 3.0 downstream, so v = 5.0 {m,s^(-1)}.

(b) Crossing time t = 60/4.0 = 15 {s}, giving a downstream drift of 3.0×15 = 45 {m}.

(c) To land opposite, the upstream component must cancel the current: 4.0sin α = 3.0, so α = 48.6° upstream of the perpendicular. The crossing speed is then 4.0cos 48.6° = 2.65 {m,s^(-1)}, and the crossing takes 22.6 {s} — longer than before. **Check:** a heading into the stream always costs time; that is the trade-off the equation makes explicit.

Solution 3 — Dot and cross products. Return →

(a) {A}·{B} = 3(-1) + 4(2) = 5.

(b) A = 5, B = √(5) = 2.24, so cos φ = 5/(5×2.236) = 0.447 and φ = 63.4°.

(c) |{A}×{B}| = |3(2)-4(-1)| = 10. Check: ABsin φ = 5×2.236×sin 63.4° = 10.0. **Check:** dot product positive means the angle is acute, and the two expressions for the cross product agree, which is the standard consistency test.

Solution 4 — Equilibrium with three forces. Return →

(a) Let the tensions be T_1 at 40° and T_2 at 60° on the other side: horizontally T_1cos 40° = T_2cos 60°, vertically T_1sin 40° + T_2sin 60° = 30.

(b) From the first, T_2 = 1.532T_1; substituting gives T_1(0.643 + 1.327) = 30, so T_1 = 15.2 {N} and T_2 = 23.3 {N}.

(c) As both angles approach 0 the vertical components can no longer support the weight, so the tensions diverge — the classic "no horizontal rope can hold a weight" limit. **Check:** the steeper wire carries more tension, which is the opposite of intuition and a good exam trap.

Solution 5 — A phasor calculation. Return →

(a) Z = 30 + 40i Ω.

(b) |Z| = 50 Ω and φ = tan ^{-1}(40/30) = 53.1°.

(c) I_0 = V_0/|Z| = 2.0 {A}, and since the reactance is inductive the current lags the voltage by 53.1°. **Check:** the numbers form a 3–4–5 triangle, the standard textbook case — and the phase angle is between 0 and 90°, as an inductive circuit must give.

Solution 6 — Superposition with phasors. Return →

(a) Two phasors of equal length at 60° add to the diagonal of a rhombus.

(b) R = 2Acos 30° = 1.732A — the same structure as the vector sum, since phasors are vectors with a phase interpretation.

(c) Intensity ∝ R^2 = $3A^2$, so the intensity is three times that of a single wave. **Check:** at 0° the factor would be 4, at 180° zero: the intensity varies from 4I_0 to 0, and 3I_0 sits between.

Solution 7 — Work as a dot product. Return →

(a) W = Fdcos 35° = 20×5.0×0.819 = 82 {J}.

(b) The normal force is N = mg - Fsin 35° = 39.2 - 11.5 = 27.7 {N}, so friction is 6.9 {N} and the work it does is -6.9×5.0 = -35 {J}.

(c) Only the component along the displacement does work, and 20cos 35° = 16.4 {N} < 20 {N}. **Check:** the net work is 82-35 = 47 {J}, which should equal the change in kinetic energy if the block accelerates from rest.

Solution 8 — A relative-velocity overtaking problem. Return →

(a) {v}_(B/A) = {v}_B - {v}_A = (-20, 15) {m,s^(-1)}.

(b) Magnitude √(400+225) = 25 {m,s^(-1)}; the direction is tan ^{-1}(15/20) = 36.9° north of west (equivalently 53.1° west of north), so B moves away from A towards the north-west.

(c) The passenger in B sees A travelling at 25 {m,s^(-1)} in the opposite direction, 36.9° south of east. **Check:** relative velocity is antisymmetric, {v}_(A/B) = -{v}_(B/A), which is a useful consistency check on any such calculation.

Solution 9 — Minimum force using components. Return →

(a) Along the plane: Wsin θ down the slope. Perpendicular: Wcos θ into the surface.

(b) Applying a force F at angle α above the plane, the normal force is N = Wcos θ - Fsin α, and the condition for no sliding is Fcos α = Wsin θ - μ N (the force helping to hold it up). Solving,

$$F = (W(\sin\theta - \mu\cos\theta))/(\cos\alpha + \mu\sin\alpha), \tag{1}$$

minimised when tan α = μ, giving F_(min) = Wsin (θ-φ) with tan φ = μ. Note the result is negative when θ < φ: the block then does not slide at all, because friction alone holds it.

(c) With μ = 0 the formula gives F_(min) = Wsin θ, applied parallel to the plane (α = 0); this is just the down-slope component of the weight, and it vanishes only when the plane is horizontal. **Check:** the tan α = μ optimum is the same "friction angle" result as the horizontal-pull problem, which is why the two are usually taught together.

Solution 10 — Complex numbers and roots of unity. Return →

(a) $|e^{i\theta}|^2 = (\cos\theta)^2+(\sin\theta)^2 = 1$, so the point lies on the unit circle for every θ.

(b) $z^3 = 1$ gives $z = e^{2\pi ik/3}$ for $k = 0, 1, 2$: that is 1, $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $-\frac{1}{2} - i\frac{\sqrt{3}}{2}$, three points at 120° intervals on the unit circle.

(c) The sum is $1 + \left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2}-i\frac{\sqrt{3}}{2}\right) = 0$. Three equal forces at 120° therefore cancel exactly — the reason a three-phase supply needs no neutral wire when the load is balanced, and the reason a tripod with symmetric legs carries a symmetric load. **Check:** the same argument with nth roots shows that n equal vectors at 360°/n intervals always sum to zero.